THE REFINERY - COHORT 8

DAGGER: Developing Airmen and Guardians with Games for Enhanced Readiness

1 PROBLEM

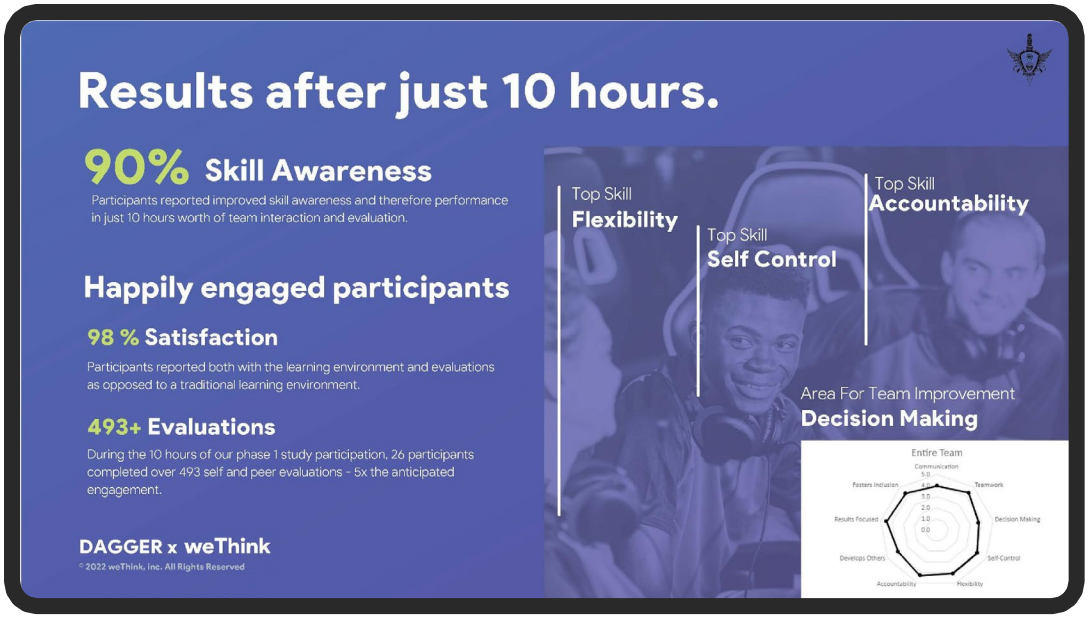
Since 2019 DAF lost more than 27 lives and over $1.43B because of poor Foundational Competencies training.

2 SOLUTION

Increase Airmen Leadership Qualities & competencies with COTS assessment tools and team-based experiential activities, just like we develop skills in real life!

3 ROI/ROX

Employees receiving competency training grew productivity 12%, with 256% ROI after 9 months.



PROJECT SUMMARY

• DAGGER, and its Department of Defense successor, Workforce Competency Development and Skills Framework, takes an existing COTS assessment tool, enabled with AI, and scaffolds it around existing occupational/experiential/team-based activity or commercial-off-the-shelf (COTS) serious games to **develop fundamental warfighter competencies** of accountability, communication decision-making, leadership and teamwork; like pilots develop competencies in flight simulators, warfighters in field exercises, and officers in a Leadership Reaction Course!

• DAGGER supports DoD Strategic Objective, SAF Imperative, DAF, CSAF and CSO strategic objectives and priorities, AETC & AU Strategic Action Plans, Key Focus Areas: Transform the Way We Learn, Develop 21st Century Warfighters, and Modernize the Learning Environment.

• Our developed product seeks customers, users and stakeholders to refine, deploy and scale DAGGER across the DoD and DAF. DAGGER can transform the way we learn, develop our 21st century warfighters, and defeat/deter our peer and near peer competitors!

AFWERX SPARK | REFINERY

- DoD Strategic Objective: Restoring the warrior ethos.
- SAF Operational Imperative 7: Readiness to Deploy and Fight.
- Department of the Air Force Strategic Objectives: Develop People.
- CSAF Action Order A: Airmen.
- CSO Guardian Ideal, Annex D: Develop and Employ Talent.

CONTACT INFO:
Matthew J. Correia
334-953-0268
matthew.correia.2@us.af.mil

COMMANDER:
Col Eugene Moore
Reserve Officer Training Corps
Maxwell AFB, AL
Holm Center (AETC)

MARKET SUMMARY

I wish I had this feedback/tool when I was a GMC (freshman and sophomore) cadet. It would have help me develop leadership competencies and show my progress.

~ AFROTC cadet

- **86% of participants** reported improvements in their competency awareness. **Cadre reported 75% of participants** demonstrated enhanced competency performance.
- Case study data showed development in the Accountability competency moving from a score of 2 to 3.5, **a 75% growth**.

DAF MARKET

DAF Total Force (Active, Guard, Reserve, Civilians): 638,087

GOVT MARKET

DoD Total Force (Active, Guard, Reserve, Civilians): 2,749,771

USG Agencies: 1,501,284

INDUSTRY MARKET

HR/Talent Management Offices

DAF TRACTION / SUCCESS

75% enhanced competency performance

Return on Investment (ROI): $204M in assets and 5 lives saved annually

Return on Experience (ROX) $205M in warfighter retention costs annually

PROJECT FUTURE

- Offering a mobile-friendly interface for real-time field assessments, enhancing accessibility and convenience during training.
- Integrating AI for analysis and coaching to boost skills, providing personalized development and actionable feedback.
- Tailoring assessments to MAJCOM contexts with specific data, ensuring relevance to operational needs.
- Refining the tool based on warfighter feedback, fostering continuous improvement and user-centric design.
- Incorporating psychometrically valid foundational competency scores, ensuring accurate and reliable assessments of critical skills.
- Benchmarking psychometrically valid scores against peer groups, enabling targeted growth through comparative insights.

TIMELINE

MAY 22

PROJECT START

Project innovator identified a pain point and worked with HAF/A1XI to find the root cause of the problem

NOV 22-JAN 23

PROTOTYPE
Successful "proof-of-concept" with 26 volunteer participants engaging in-person and remotely with application.

SEP 23-SEP 24

DESIGN
Successful design, test and iteration with 1.5k AFROTC and OTS volunteer participants engaging in-person and remotely with application.

SEP 24-SEP 25

DEVELOPMENT
Currently iterating design to incorporate AI to curate feedback, enable application for mobile devices enabling foundational competencies assessment anytime, anywhere, in any environment (classroom, operational, social).

OUR TEAM

Matthew Correia, Civ, DAF

Joshua Hayden, Civ, DAF

Ms. Katrina Salazar, CEO, weThink

Mr. Nick Potter, CTO, weThink

Photo unavailable

Adam Brewbaker, TSgt, USAF